ACEFA
AUSTRALIA-AOTEAROA CONSORTIUM FOR EPIDEMIC FORECASTING & ANALYTICS

Lecture 4
Analysis of epidemic time-series part 1: quantifying temporal trends

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Prepared for the ACEFA EpiAnalytics Course at Australian CDC, Canberra, November 2026
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What is an epidemic time series?

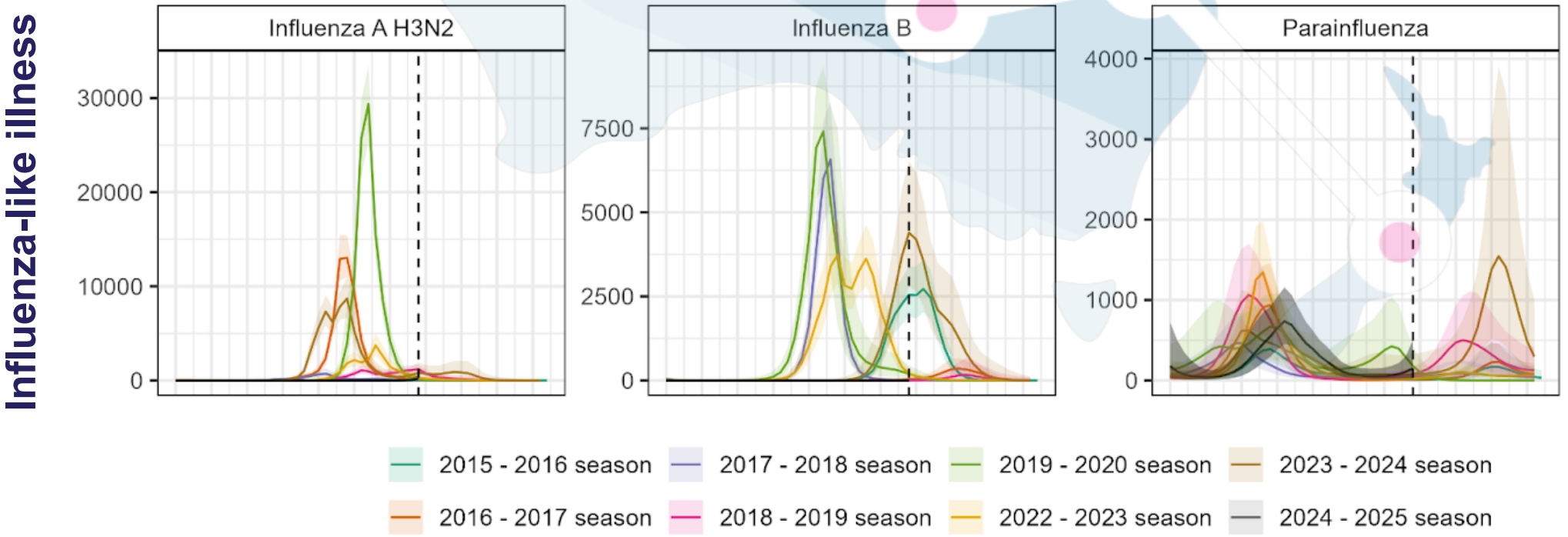
- A quantity measured at successive equally spaced points in time.
  - Could be measured: daily, weekly, monthly, etc
  - Value at current time point depends on previous time points. i.e. data points are not independent

- Examples quantities include:
  - Cases
  - Hospitalisations
  - Deaths

- Note that the time series of infections is rarely measured

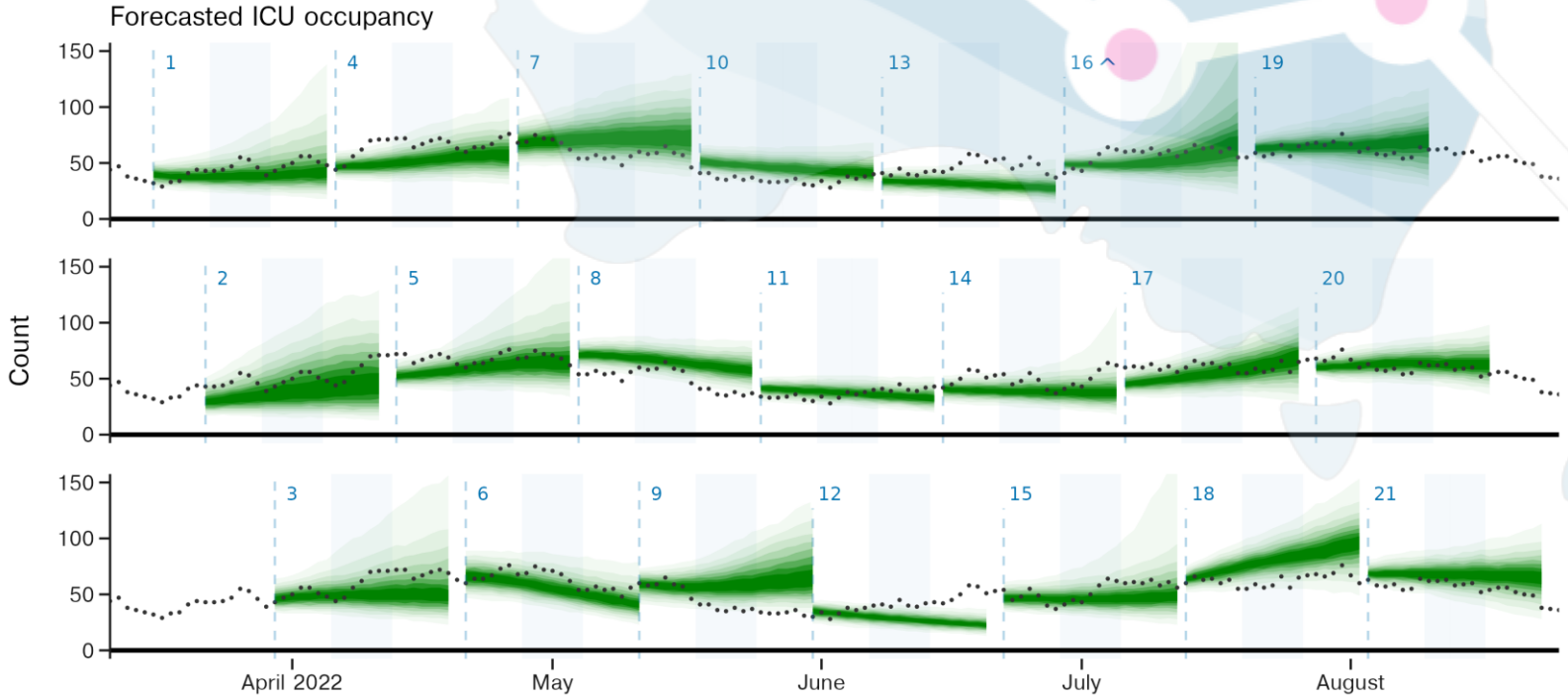
Why do we analyse epidemic time series?

- By analysing past and current trends, we can predict future trends
- For example:
  - Analysing past seasonal dynamics of respiratory pathogens can inform predictions of onset, peak timing, and peak burden for future seasons



Why do we analyse epidemic time series?

- By analysing past and current trends, we can predict future trends
- For example:
  - Current trends can be indicative of short-term future trends



Tobin et al. 2025. Communications Medicine

This lecture's scope



In this lecture we will:

- Describe why transmission can change over time and why it is important to monitor
- Present two mathematical formulations for quantifying these changes: the effective reproduction number and the epidemic growth rate
- Relate these quantities to an exemplar time series of infections

We will not cover:

- "Noisy" or delayed time series data — this will be covered in lecture 5 (analysis of epidemic time-series part II)
- Handling of epidemic time series comprised of signals from multiple pathogens — this will be covered in lecture 6 (analysis of epidemic time-series part III)

The effective reproduction number, R_e

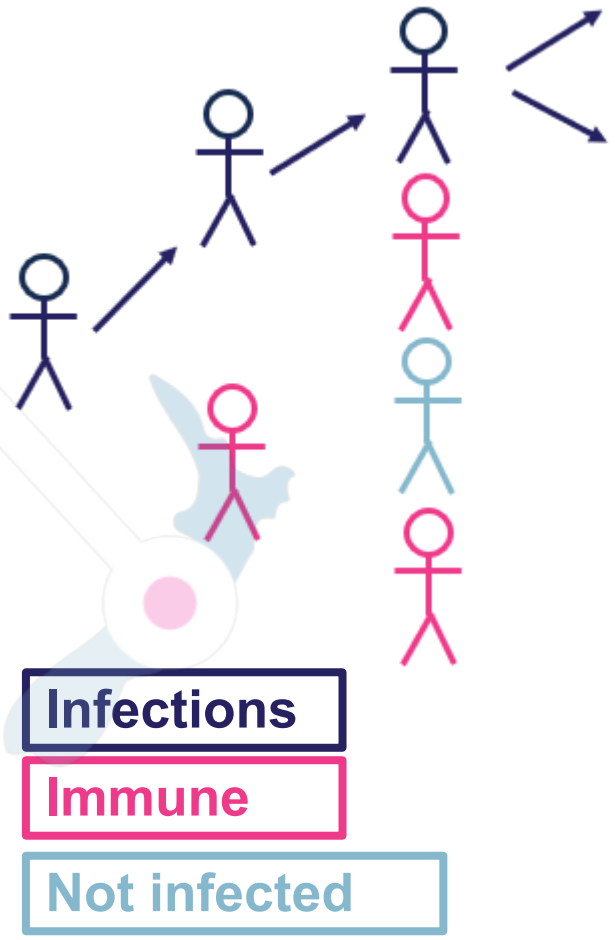
The effective reproduction number, R_e

The basic reproduction number:

- The expected number of secondary infections generated by one infected individual in a completely susceptible population

The effective reproduction number

- The expected number of secondary infections generated by an infected individual in a population which may not be entirely susceptible
- Also called the time-varying reproduction number, R_t
- Informs the intensity of interventions currently required for disease control
  - When $R_e < 1$ infection incidence is decreasing
  - When $R_e > 1$ infection incidence is increasing
  - When $R_e \approx 1$ infection incidence is stable



Why does R_e change through time?



- Changes in the susceptibility of the population
  - Decreases in susceptibility due to infection- and vaccine-induced immunity
  - Increases in susceptibility due to waning immunity or new susceptibles (e.g. births)

- Changes in the infectiousness of the pathogen
  - Decreases in infectiousness due to infection- and vaccine-induced immunity
  - Decreases in infectiousness due to antivirals
  - New variants with altered transmissibility

- Changes in contact rates and other behaviours
  - Lockdowns can reduce all contact
  - School or workplace closures will change contact rates for subsets of the population
  - Isolation and quarantine can reduce contact between infectious and susceptible individuals

How do we calculate $R_e(t)$?

- We can calculate $R_e(t)$ using the equation:

Effective reproduction number

$$R_e(t) = \frac{I(t)}{\sum_{\tau=0}^{n} I(t-\tau)g(\tau)}$$

Infection incidence on the current day

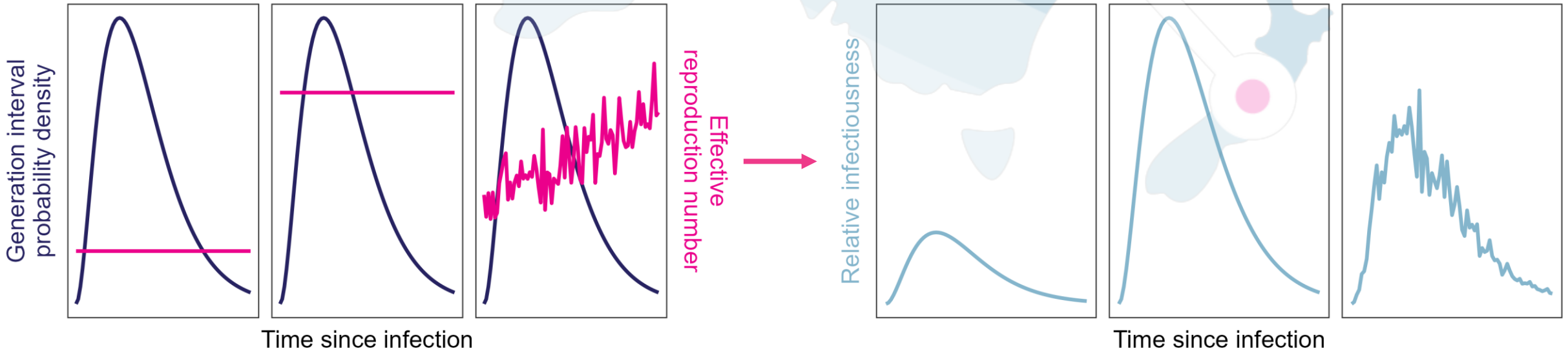
GI distribution

Infections incidence on previous days

Infection incidence, the generation interval and R_e

- The generation interval (GI), describes the time between primary and secondary infections when $R_e\ (t)$ is constant
  - The GI describes the infectiousness of an individual (or population average) over time following infection, $g(\tau)$
- R_e describes the expected number of secondary infections generated by an infected individual
- The rate of secondary infections at time τ since infection at time t is given by:

$$g(\tau)R_e(t+\tau)$$



Infection incidence, the generation interval and R_e

- We can quantify how infections in the past gave rise to infections at the current time point.

Infection incidence on current day

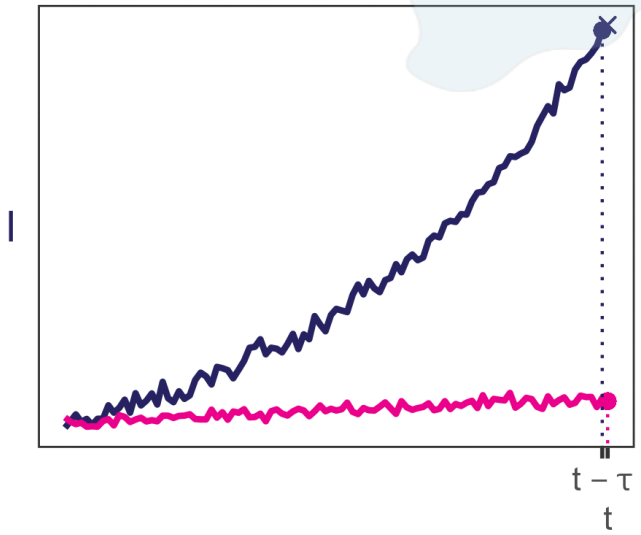
GI distribution

Effective reproduction number

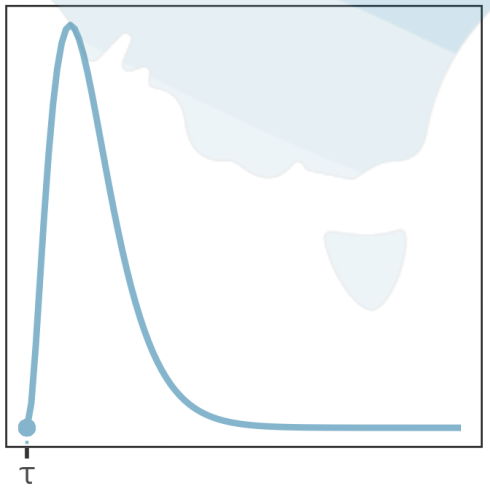
$$I(t) = \sum_{\tau=0}^{\infty} I(t-\tau)\, g(\tau)\, R_e(t)$$

$$I(t) = R_e(t) \sum_{\tau=0}^{\infty} I(t-\tau)\, g(\tau)$$

Infections incidence on previous days



R_e GI



Contribution to $I(t)$:

$$I(t-\tau)\, g(\tau) R_e(t)$$

How do we estimate $R_e(t)$?

- If we have a perfect time series of the infection incidence and an estimate for the generation interval, then we can straightforwardly calculate $R_e(t)$:

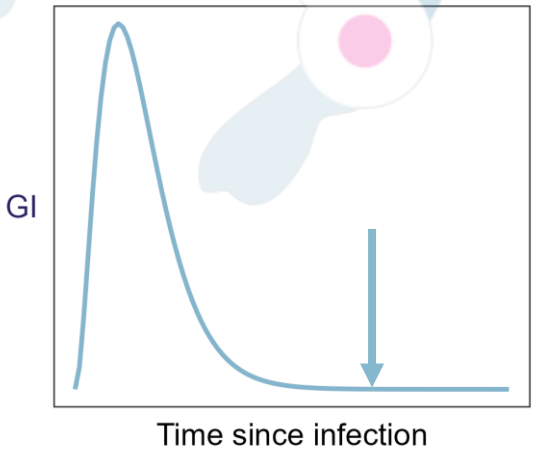
Effective reproduction number

$$R_e(t) = \frac{I(t)}{\sum_{\tau=0}^{n} I(t-\tau)g(\tau)}$$

Infection incidence on current day

GI distribution

Infection incidence on previous days



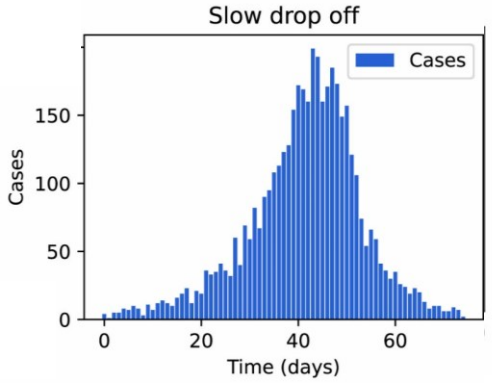
  - We perform the summation up to some limit n for computational feasibility (value at which $g(n) \approx 0$)

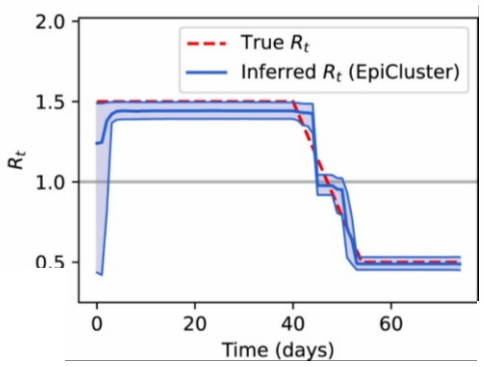
How do we estimate $R_e(t)$?

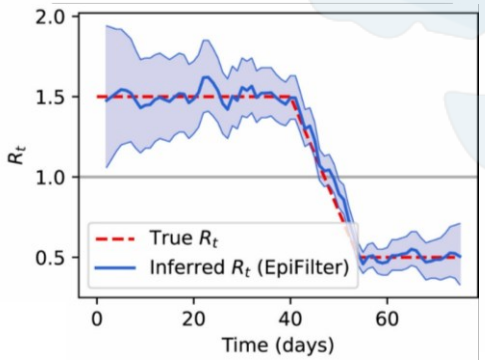
- In general, we cannot straightforwardly calculate, $R_e(t)$ from data
  - Transmission and observation processes give rise to measurement noise
  - We will cover biases in the observation process in lecture 5

- We can estimate $R_e(t)$ using statistical models that account for noise, but they require assumptions. For example:
  - Assume that $R_e(t)$ changes smoothly?
  - Assume that $R_e(t)$ can occasionally change sharply?

- Need to be wary of these assumptions when interpreting estimates







Creswell et al. 2023. Journal of Theoretical Biology

How do we estimate $R_e(t)$?

What if we don't know the generation interval?

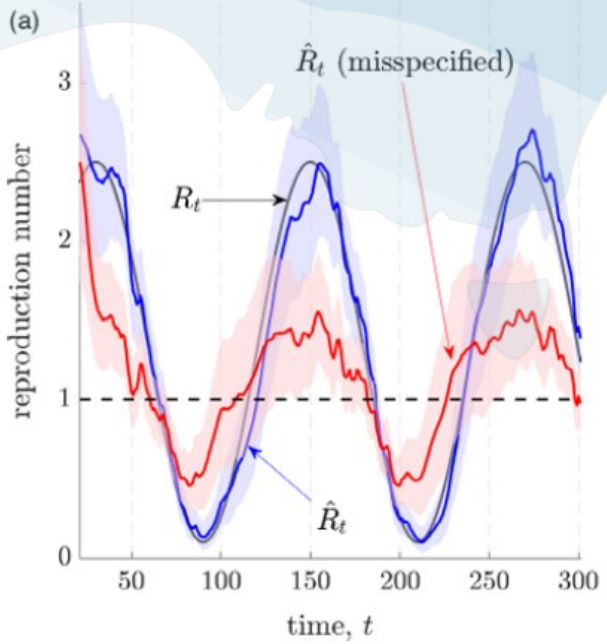
- Calculating $R_e(t)$ requires an estimate of the GI distribution

$$R_e(t) = \frac{I(t)}{\sum_{\tau=0}^{n} I(t-\tau)g(\tau)}$$

- Using a biased estimate of the GI distribution gives biased estimates of $R_e(t)$

$R_e(t)$ inferred from simulated Ebola outbreak data

- Simulated $R_e(t)$
- Inferred $R_e(t)$ - correct GI assumed
- Inferred $R_e(t)$ incorrect GI assumed



Parag et al. 2022. J R Stat Soc Ser A Stat Soc

How do we estimate $R_e(t)$?



What if we don't know the generation interval?

- GIs are hard to estimate
  - Infection times are not directly observable
  - Measured GI (as opposed to the intrinsic GI) will change through time
  - GI estimates may have substantial uncertainty – rarely included in uncertainty of $R_e(t)$ estimates
  - GI estimates can be biased (i.e. methods not accounting for data observation biases)
  - The intrinsic GI can change (e.g. variants of COVID-19)

- Always be discerning when presented with estimates of $R_e(t)$
  - What methodological assumptions were made?
  - What GI distribution was assumed?
  - How might estimates change under different assumptions?

The epidemic growth rate, r

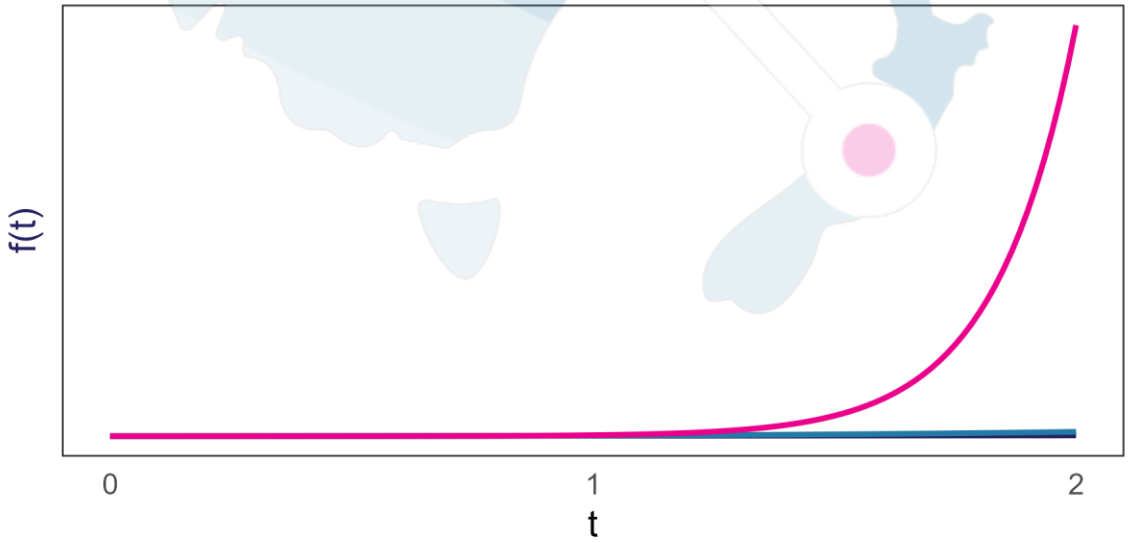
What is exponential growth?

- Exponential growth occurs when a quantity grows at a rate proportional to its current size

$$\frac{df(t)}{dt} = rf(t), \qquad f(t) = f(0)e^{rt}$$

- For example, when the quantity is two times its current size it will also grow at twice its current rate.

Linear growth: $f(t) = 25 \times t$
Cubic growth: $f(t) = t^3$
Exponential growth: $f(t) = e^t$



What is exponential growth?

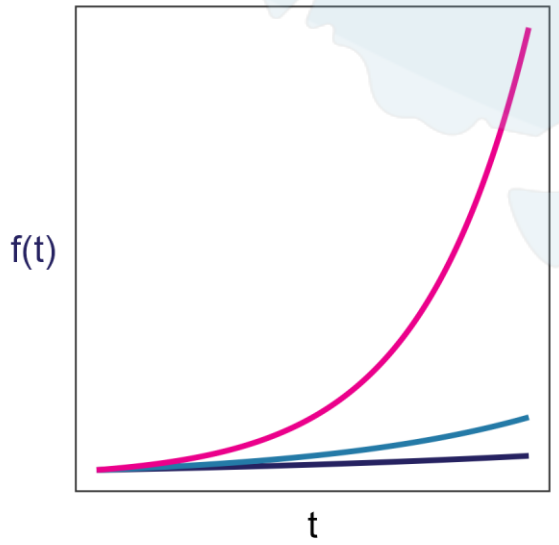
- Exponential growth is linear on a log-scale with the gradient given by the growth rate r
  - When $r < 0$ the quantity decreases
  - When $r > 0$ the quantity increases

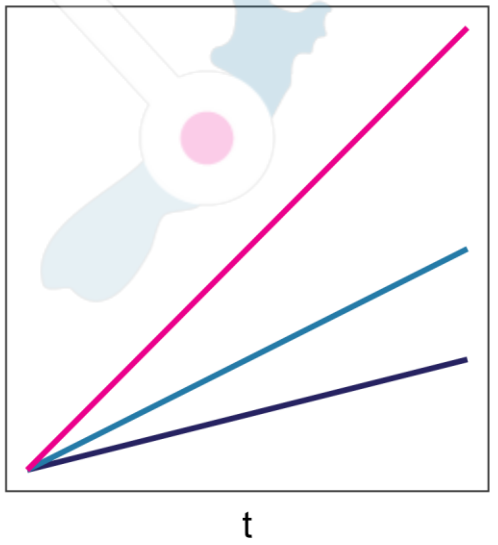
- The doubling (halving) time is the time taken for the quantity to double (halve) in value and is given by:

$$doubling\ time = \frac{log(2)}{r}$$

$f(t) = e^{r \times t}$

$r = 0.05$: ~14 day doubling time
$r = 0.1$: ~7 day doubling time
$r = 0.2$: ~3.5 day doubling time

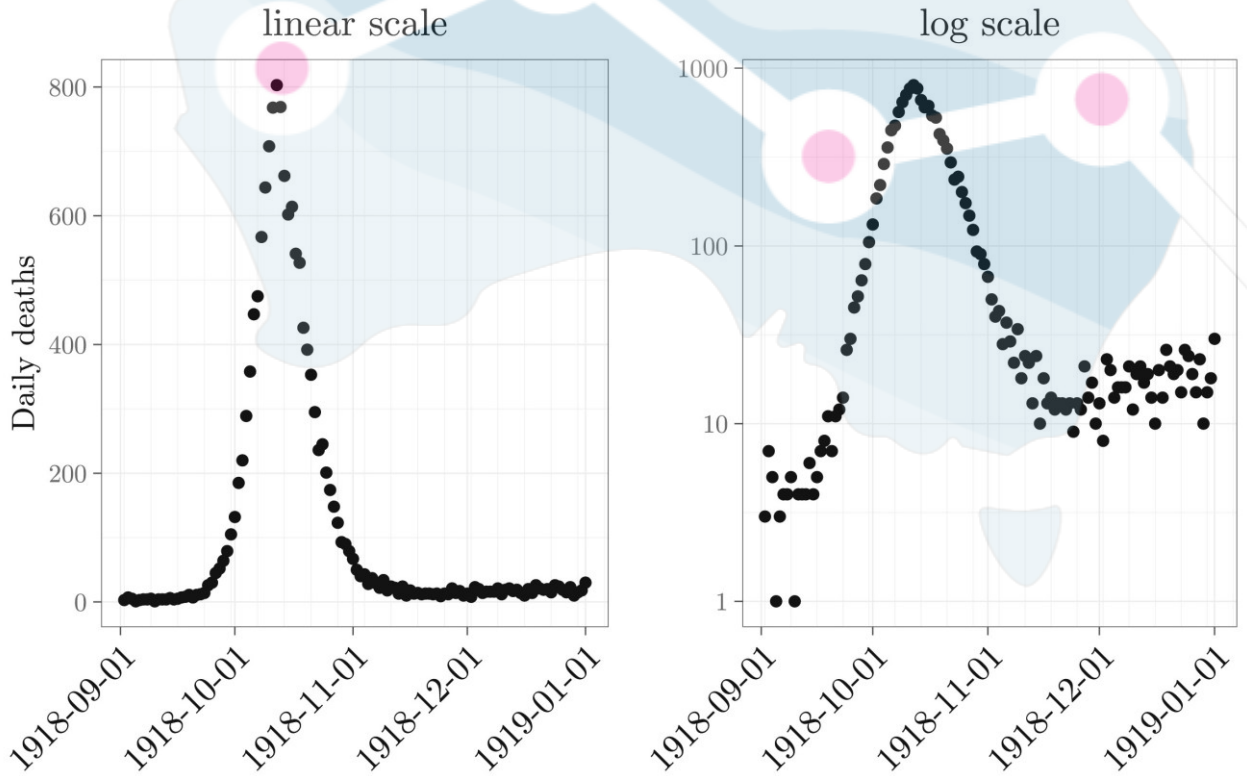




Why is exponential growth relevant to epidemics?

- **Epidemics often exhibit sustained periods of constant exponential growth/decline:**

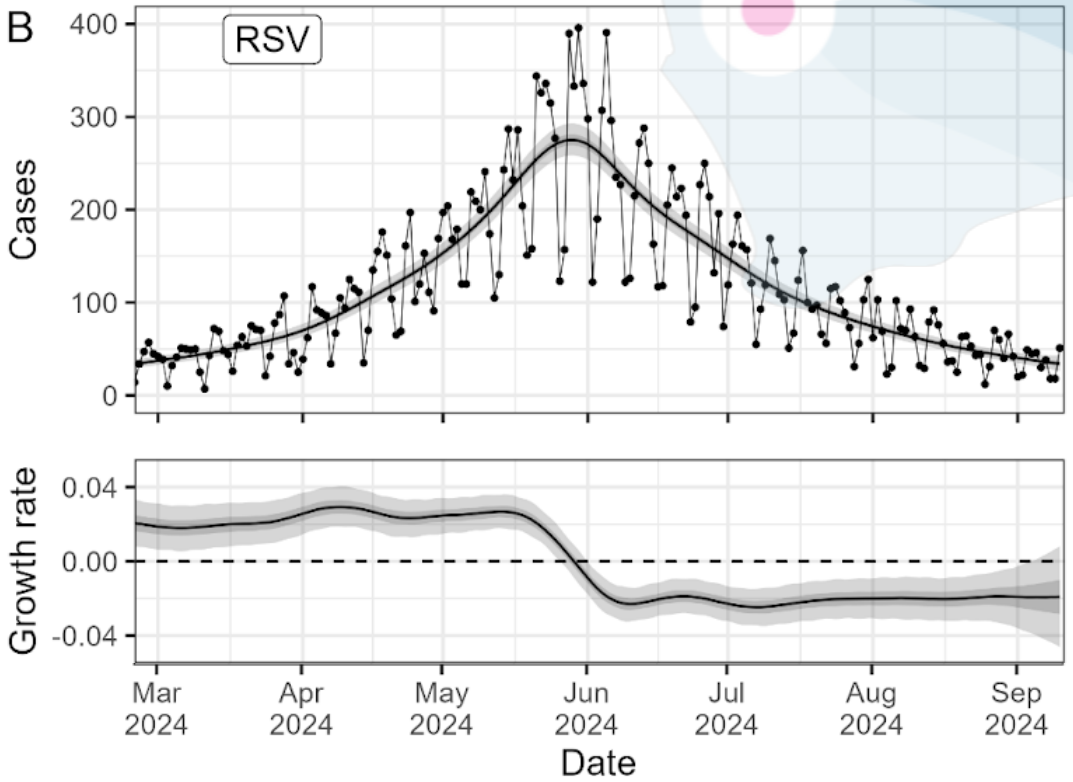
Influenza Deaths in Philadelphia during the 1918 pandemic

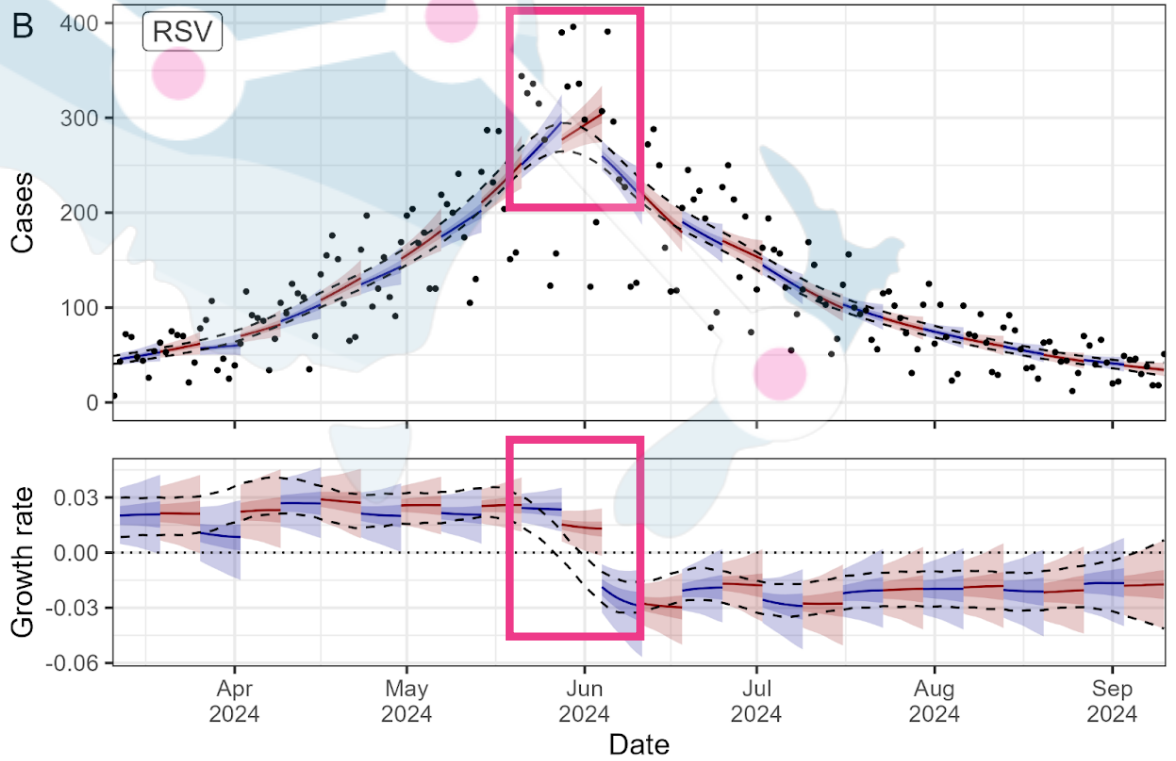


Ma 2020. Infectious Disease Modelling

Infection incidence and the epidemic growth rate

- The growth rate will change over time
- The growth rate will change fastest at the epidemic peak from above to below 0
  - This can lead to overestimates of future infection levels when forecasts are made immediately prior to epidemic peaks





Eales 2025. Communicable Diseases Intelligence

Infection incidence and the epidemic growth rate

- We can write the rate of change of infection incidence in terms of a time-varying growth rate:

$$\frac{dI(t)}{dt} = r(t)I(t)$$

- This then has the solution:

$$I(t) = I(0)e^{\int_0^t r(\theta)d\theta}$$

- Or

$$I(t) = I(t-1)e^{r(t)}$$

Infection incidence at current time

Infections incidence at previous time

Growth rate from previous time to current time

How do we estimate the epidemic growth rate?

- We have the equation linking infection incidence and growth rate:

$$I(t) = I(t-1)e^{r(t)}$$

- We can then calculate the growth rate from the equation:

Epidemic growth rate →

$$r(t) = log\big(I(t)\big) - log(I(t-1))$$

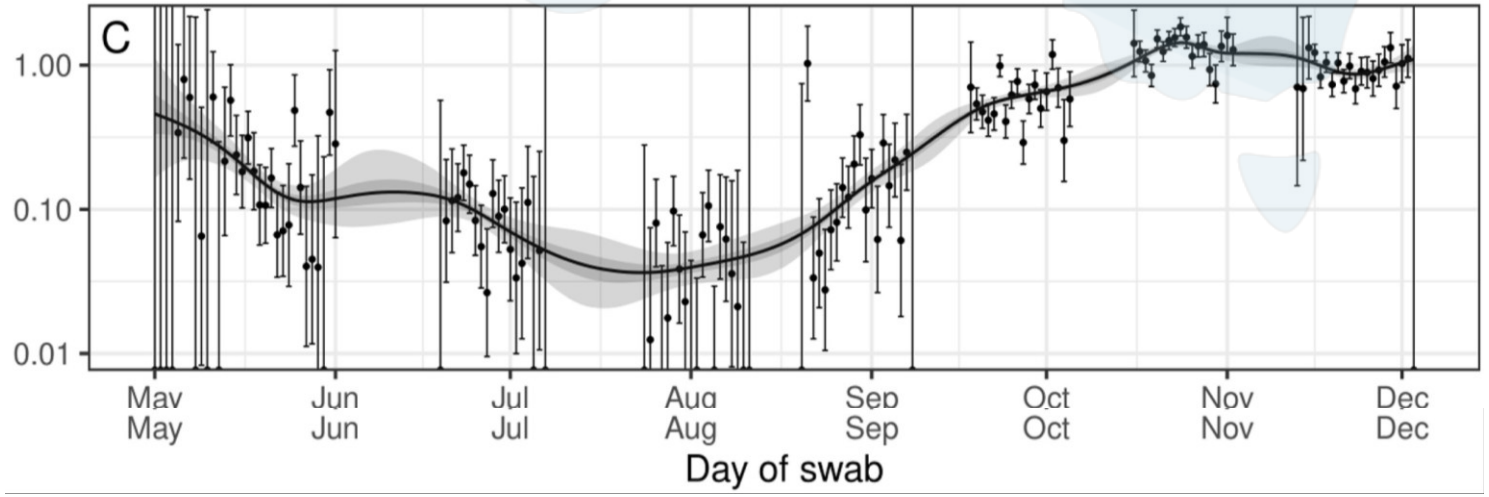
Difference between log of infection incidence at current time and previous time step

- The units of $\mathrm{r}(t)$ will be t^{-1} (e.g. week^{-1}, day^{-1})

How do we estimate the epidemic growth rate?

- In general, we cannot straightforwardly calculate, $r(t)$:
  - The time-series of infection incidence will exhibit noise
  - We will cover biases in the observation process in lecture 5

- We can estimate $r(t)$ using statistical models that account for noise but must make assumptions. For example:
  - Assume that $r(t)$changes smoothly?
  - Assume that $r(t)$ occasionally changes sharply?
  - Assume that $r(t)$ is constant for a fixed period
- Need to be wary of these assumptions when interpreting estimates



Eales et al. 2022. Epidemics

Should we report $R_e(t)$ or $r(t)$?

The relationship between $R_e(t)$ and $r(t)$

- A time-series of infection incidence can be written in terms of:
  - The effective reproduction number:

$$I(t) = R_e(t) \int_0^\infty I(t-\tau) g(\tau) d\tau$$

  - The time-varying epidemic growth rate:

$$I(t) = I(t-\tau)\, e^{\int_{t-\tau}^{t} r(\theta) d\theta}$$

- Can obtain a general relationship between $R_e(t)$ and $r(t)$

$$\frac{1}{R_e(t)} = \int_0^\infty e^{-\int_{t-\tau}^{t} r(\theta) d\theta} g(\tau) d\tau$$

e.g. when generation interval is discrete

$$g(\tau) = 1, \quad \tau = \overline{g}$$
$$g(\tau) = 0, \quad \tau \neq \overline{g}$$

$$R_e = 1 + \overline{g} \times r$$

- Can obtain a simple formula when $R_e(t)$ and $r(t)$ are constant, for different parametric forms of $g(\tau)$. For more details, please refer to: Wallinga and Lipsitch 2007 Royal Society Proc Biol Sci.

$R_e(t)$ vs $r(t)$: which is better to report?



- We have presented two formulation for quantifying transmission
  - The effective reproduction number, $R_e(t)$
  - The epidemic growth rate, $r(t)$

$R_e(t)$:

- Can be more informative on transmission dynamics (i.e. number of secondary infections per individual) and effect of interventions
- May be biased by incorrect estimates of or assumptions on the GI

$r(t)$:

- Does not require the GI to be specified
- More naturally quantifies the epidemic speed (i.e. doubling time calculated directly from $r(t)$)

- Report both and be discerning of how they were estimated (i.e. the method used)
  - What assumptions were made?
  - Were the assumptions reasonable for the epidemiological context?
  - How sensitive are the estimates to these assumptions?

Practical

In today's practical we will:

- Estimate $R_e(t)$ and $r(t)$ using a statistical model (implemented in EpiNow2)
- Compare estimates of $R_e(t)$ and $r(t)$ made:
  - Using different assumed GI distributions
  - Using different datasets (more detail in following lecture)
  - Using different datasets correcting for delays (more detail in following lecture)

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